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respectively, with the help of DIP-SW1. Q0 through Q9 outputs of IC3 are used for device selection from 0 to 9, respectively, with the help of DIP-SW2.

Lock circuit (shown inside the dotted line) is built around flip-flop CD4013, two AND gates, a transistor and relay RL1. If Vx and 'on' inputs of AND gate N3 are high, its output sets flip-flop CD4013. The high output of the flip-flop drives the transistor into saturation and relay RL1 energises to turn the device 'on'. If Vx and 'off' inputs of AND gate N4 are high, its output resets flip-flop CD4013. As a result, relay RL1 de-energises to turn the device 'off'.

The 230V, 50Hz AC mains is stepped down by transformer X1 to deliver a secondary output of 9V-0-9V, 500mA. The transformer output is rectified by a full-wave rectifier comprising D1 and D2, filtered by capacitor C1 and regulated by IC 7806 (IC5), which provides a 6V DC output. Capacitor C2 provide further filtering.

Construction

An actual-size, single-side PCB for the computerised electronic lock is shown in Fig. 3 and its component layout in Fig. 4. Assemble the circuit on a printed circuit board (PCB) to minimise time and assembly errors. Carefully mount the components and double-check for any overlooked error. On the PCB, provide connectors for 'on', 'off' and 'Vx' points. The clock circuit is separate for every device, and fitted along with the device.

If you want to switch on any device, say, device No. 13, connect the clock circuit to 'on', 'off' and 'Vx' points and close switch S1 of DIP-SW1 and switch S2 of DIP-SW2. Now log into the software and enter the device No. (13) in the device number box and click 'on' button. The flip-flop is set and relay RL1 energises to activate the device.

Software

The source program for the electronic lock is written in C++ and compiled using Turbo C++. It is well commented and easy to understand.

When you click 'clock.exe', a screen pops up that prompts you to enter the password. Type the password as 'lock' and press 'enter' from the keyboard. The main screen appears. Click the control panel button to get the control panel screen as shown in Fig. 2. Now you can control a device by entering the device number and clicking 'on' or 'off'.

In 'select all' mode, you can control up to 20 devices through this program. The number of devices to be controlled can be set only from the program.

In 'group' mode, you can create groups of different devices (like a group of device numbers 2, 4, 5 and 8). For every group, one '.dat' file is created with the group name. Inside the .dat file, the device number ends with '-1'. You can add up to 99 devices to a single group. ●

EFY note. All the relevant software files of this project have been included in this month's EFY-CD.